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An efficient automatic arousals detection algorithm in single channel EEG

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ABSTRACT

Background and objective: Electroencephalographic arousal is a transient waveform that instantaneously happens in sleep as an inherent component. It has distinctive amplitude and frequency features. However, it is visually difficult to distinguish arousal from the background of the electroencephalogram. This visual scoring is important for brain researches, sleep studies, sleep stage scorings and assessment of sleep disorders. The scoring process is a time-consuming and difficult clinical procedure which is evaluated by sleep experts. It may also have subjective consequences due to the variability of personal expertise of physicians. Conversely, this scoring process can be significantly accelerated with computer-aided automated algorithms. Moreover, reproducible and objective results can be obtained. In this work, we propose a novel algorithm for the automatic detection of electroencephalographic arousals in sleep polysomnographic recordings.

Methods: The approach uses a well-known time-frequency localization method, the continuous wavelet transform, to identify relevant arousal patterns. Special emphasis was carried out to produce a robust, reliable, fast and artifact tolerant algorithm. In the first part, the electroencephalographic scalogram, the squared magnitude of the continuous wavelet transform, was obtained. The mean and variance of the scalogram coefficients were determined as novel features. Support vector machine was applied as a classifier. Half of the recordings were used for training with five-fold cross-validation and a high accuracy training rate was obtained. Then, the rest of the recordings were used for testing.

Results: As a result, the overall sensitivity, specificity, accuracy, and positive predictive value of the algorithm are 94.67%, 99.33%, 98.2%, and 97.93%, respectively.

Conclusion: In this paper, we have shown that the electroencephalographic arousal pattern can be characterized by the scalogram in the wavelet domain. The proposed algorithm works with high accuracy, reproducibility and gives objective results without case-specific sensitivity.

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1. Introduction

Nowadays, one of the most important problems of society is sleep disorders, which affect an individual's health, safety, and quality of life. Daytime sleepiness is a condition seen in nearly everyone. Especially for some people, excessive sleepiness actually gets in the way of daily work, social life, traffic, and even leisure activities. In addition, recurrent sleepiness makes people want to nap repeatedly, even at work. The most common reason of daytime sleepiness is the interruption in night sleep caused by a short stimulus, which is called electroencephalographic (EEG) arousal [1,2]. According to the American Academy of Sleep Medicine (AASM), an EEG arousal is an abrupt shift in the EEG frequency including alpha, theta and/or frequencies greater than 16 Hz (but not spindles), that lasts at least 3 s with at least 10 s of previous stable sleep [3].

Polysomnography (PSG) is the gold standard clinical technique for the assessment of sleep disorders [4]. The occurrence of EEG arousals in PSG is important for a good assessment of the patient's sleep continuity. In the clinic, these records are visually examined by an expert sleep physician for the diagnosis. Visual scoring of arousals in a whole night EEG signal is time-consuming, difficult, and requires personal expertise [1]. It may also have subjective results due to high inter-scorer variability [5,6]. Hence, the development of computer-based decision support systems has become more important in recent years.

There have been few studies on the automated detection of the arousal in the literature for recent years. Drinnan et al. presented an explorative study in which the frequency and

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activity changes of EEG were evaluated [5]. Later, De Carli et al. described a method for the automatic detection of arousal based on wavelet analysis [7]. Agarwal presented a method which was based on adaptive segmentation and spectral feature extraction for automatic detection of arousal [8]. Cho et al. proposed an automated arousal detection algorithm based on the time-frequency analysis [9]. Shmiel et al. presented a different approach based on data mining techniques for automatic detection of arousal [10]. Sugi et al. proposed a method for automatic detection of EEG arousal in patients who are suffering from sleep apnea [11]. Álvarez-Éstevez et al. presented a method for the detection of arousal using two EEG channels and the electromyography (EMG) [1]. Behera et al. presented an automatic arousal detection methodology. According to this study, arousal events can be detected where the power spectral density of the filtered signal is over a threshold value [12]. Wallant et al. proposed a method for the automatic detection of artifacts, which contains a module for the characterization of EEG arousals [13]. Fernández-Varela et al. presented a multi-channel analysis which starts with signal conditioning and then gets an initial set of EEG arousals based on frequency measures over the signals [14]. Finally, in another study of Fernández-Varela et al., a method was described for the automatic detection of arousal based on Hjorth's parameters and the sleep stage [15].

This work proposes a novel method for automatic arousal detection from a single channel EEG. The algorithm is based on continuous wavelet transform (CWT) and does not require any preprocessing step. In this paper, we introduce that the EEG arousal pattern can be characterized by the scalogram in the wavelet domain. The goal of the study is to fulfill two main requirements, robustness and accuracy, with respect to different cases/patients in an efficient and genuine way.

This paper is an extension of the work presented at SIU 2017 [16]. The main enhancements are the application of the linear support vector machine (SVM) for classification, and an extended data set for validation. The rest of this paper is organized as follows. Section 2 begins with the description of the database and the properties of the signals. Then, the developed algorithm is depicted with a block diagram. In this context, the methodological approach of the research is explained. Moreover, proper way to formulate the problem is presented. The obtained novel features are also introduced. This section is completed by explaining the mathematical basis and the properties of the classifier. The training and test results of the classifier are given in Section 3, which is followed by Section 4 with a discussion on the extensive results. Automatic arousal detection algorithms presented in the literature and the proposed solutions by the developed method are compared in details. Finally, we present some concluding remarks in Section 5.

2. Materials and methods

2.1. The database and PSG recordings

In the present study, a set of overnight PSG data was used. The recordings were taken from the Sleep Heart Health Study (SHHS) PSG database of National Sleep Research Resource (NSRR). SHHS is a multi-center cohort study supplied by the National Heart Lung & Blood Institute to determine the cardiovascular and other consequences of sleep-disordered breathing [17-19]. All night PSG recordings contain oxygen saturation (SaO_2), pulse rate (PR), EEG (C3/A2), electrocardiogram (ECG), EMG, right and left electrooculogram (EOG), EEG (C4/A1), airflow, thoracic respiration and abdominal respiration, body position, ambient light, oximetry. In this study, we only used one channel EEG data of each PSG record. The EEG signals were recorded with 125 samples per second with 8-bit resolution. C3/A2 derivation of EEG was used for the detection of EEG arousal. It is stated that there is no soft pre-processing on the signal received from the database. However, 0.15 Hz high-pass filter was applied to the signal using embedded hardware as a standard tool in the data acquisition system of PSG.

The PSG recordings have annotations of the arousal events marked by the expert sleep physician. All records in the database are in European data format (EDF) format. The EDF file can be opened with an open source EDF viewer given in website (https://sleepdata.org/). It was developed by a collaboration between Case Western Reserve University and Brigham and Women's Hospital for the extension of the original Physio-MIMI viewer (https://sleepdata.org/tools). A typical PSG record with all channels is displayed in Fig. 1.

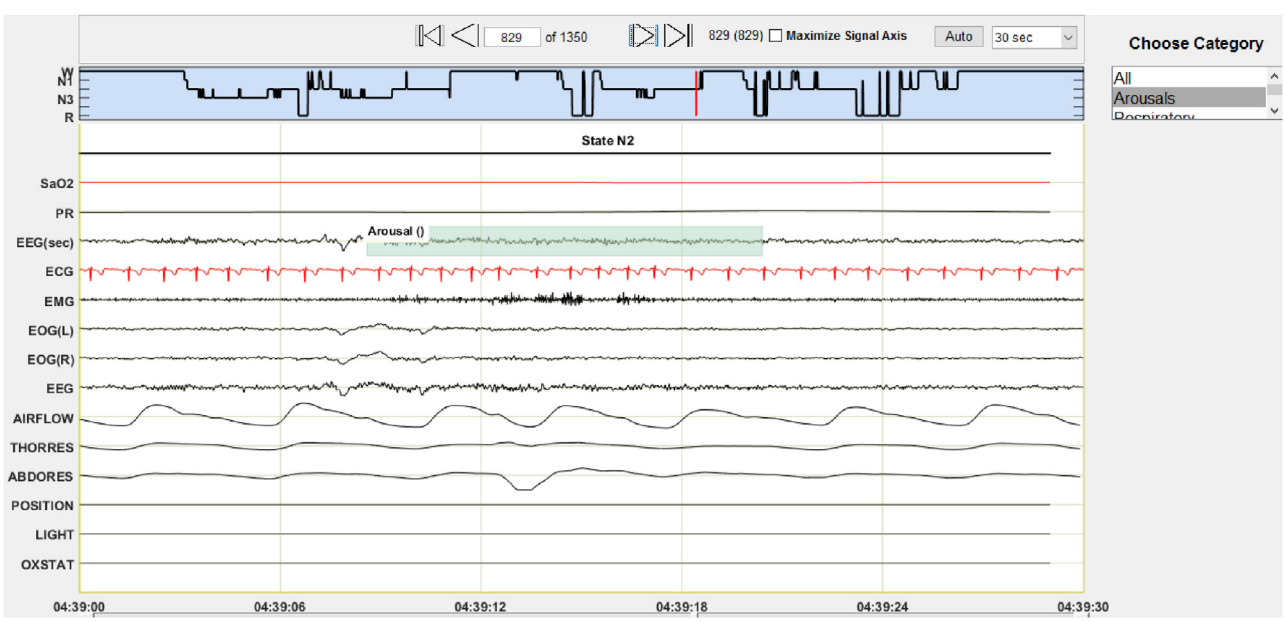

Fig. 1. A sample EDF file opened with the Physio-MIMI viewer. All signals in the PSG recordings and sleep score annotation can be seen (https://sleepdata.org/tools).

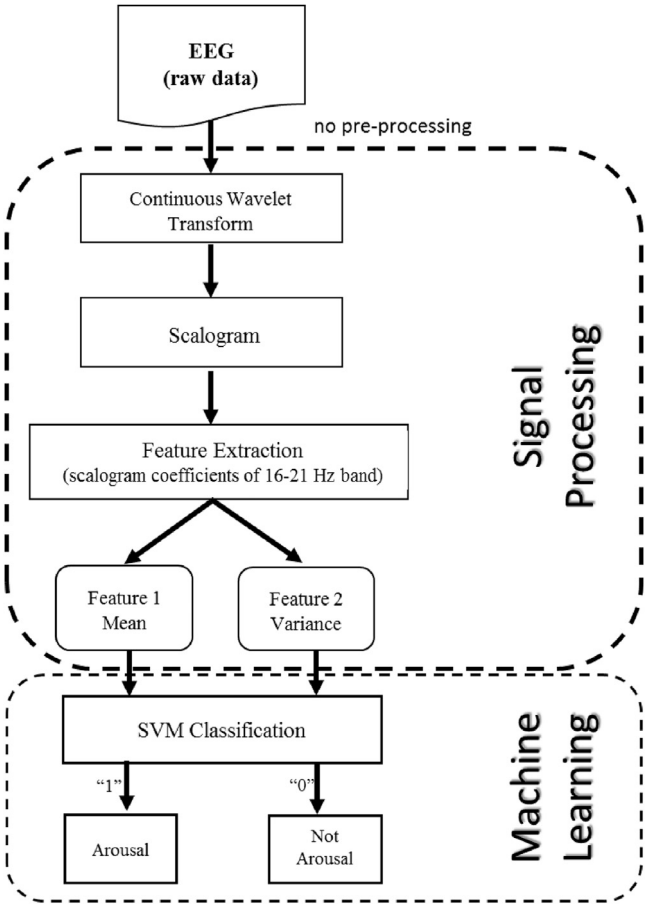

Fig. 2. Flowchart of the proposed algorithm.

2.2. Proposed algorithm

Following the design depicted in Fig. 2, we propose a multistage method which is based on the extraction of the time-frequency characteristics of the EEG arousal from the raw signal without signal conditioning. Also, the proposed method carries out a classification using machine learning algorithm. The first stage includes the feature extraction after obtaining the scalogram coefficients of the raw EEG signal. The second stage involves the formation of the training and test sets, followed by the classification process. The methods used in each part of the proposed algorithm are explained in the following subsections. This algorithm was developed using MATLAB 2015b, The Math Works Incorporation.

2.3. Time and frequency characteristics of EEG

Classical or modern spectral analysis techniques ensure a complete and proper solution for stationary signals. However, many signals, especially those of physiological origin, are non-stationary and these signals vary significantly in their properties over time [20]. For instance, the time and frequency characteristics of the EEG signal vary considerably depending on the electrical activity of the brain. Moreover, these changes that occur instantaneously in the signal are often of primary interest.

Although there is no time localization, Fourier transform (FT) provides a decomposition of a signal into individual frequency components and establishes the relative intensity of each component, for the frequency domain analysis [21]. Time-domain knowledge is highly important in many non-stationary signals such as EEG. A wide variety of methodologies have been developed to accomplish both time and frequency knowledge from a signal. Fundamentally, they can be branched into two groups: time-frequency and time-scale methods.

In order to solve the time localization problem, short-time Fourier transform (STFT) has been developed. STFT is based on a simple approach of slicing the waveform into a number of short segments and performing the analysis on each of these segments using the standard FT. The spectral content of many physiological signals, such as EEG, changes very quickly, so finding an appropriate window length is problematic. Decreasing the window length to improve time resolution will reduce frequency resolution and vice versa. Short window length causes loss of low-frequency components, whereas long window length provides insufficient time resolution. Hence there is an inherent trade-off between time and frequency resolution due to the uncertainty principle [21]. Although this method has been successfully applied to time-varying signals in a number of biomedical applications, the time-frequency balance and the optimal window length selection still remain to be two main problems of the spectrogram.

The inadequacy of the FT to recognize both the time and frequency properties of the waveform and the disadvantage of the fixed window length used in the STFT led to the development of an alternative approach [20]. The wavelet transform, introduced at the beginning of the 1980s by Morlet et al. [22], can be used as yet another way to characterize the properties of a waveform that changes over time [20]. However, in this case, the waveform is divided into segments of scale instead of time. This approach gives result in a good frequency resolution and poor time localization for low-frequencies. Conversely, the frequency resolution is reduced with better time localization.

Discrete wavelet transform (DWT) has excellent signal decomposition property for signals. Since DWT is a very efficient calculation tool, it has been applied to almost all fields of signal processing including signal decomposition, compression, denoising, numerical integration, and pattern recognition [23]. On the other hand, the CWT finds most of its applications in time-varying signals analysis, where it yields an affine invariant time-frequency representation [24]. Due to this particular approach and the superiority of the time-frequency characterization of non-stationary signals, the CWT is used in the presented work.

2.3.1. The continuous wavelet transform

A wavelet family can be composed of a function $\psi(t)$, called "Mother wavelet", which is constrained in a limited interval to be more flexible in extracting time and frequency knowledge. Daughter Wavelets, $\psi_{b,a}(t)$, are constructed by translation with a factor "b" and dilation with a scale parameter "a" [23], defined as:

$$\psi_{b,a}(t) = |a|^{-\frac{1}{2}}\,\psi\left(\frac{t-b}{a}\right) \tag{1}$$

The wavelet transform basically decomposes the signal over dilated and translated wavelets. This difference makes it possible to perform a multi-resolution analysis with wavelet transform. Therefore, the main wavelet to be used must be selected in order to calculate the CWT as the following inner product:

$$W_x(b,a) = \int_{-\infty}^{\infty} x(t)\psi^*_{b,a}(t)dt \tag{2}$$

where $x(t)$ is the original signal, $\psi(t)$ is the Mother wavelet function and the superscript $*$ denotes the complex conjugate [23].

In Fig. 3a, 30 s of EEG signal from the C3/A2 channel that includes arousal is shown. This arousal and the other arousals were visually identified by a sleep expert, and all scored data were given in the database. Then, the CWT is applied to this epoch using complex Morlet wavelet according to the results of the previous study in which the scalogram was analyzed as an instantaneous

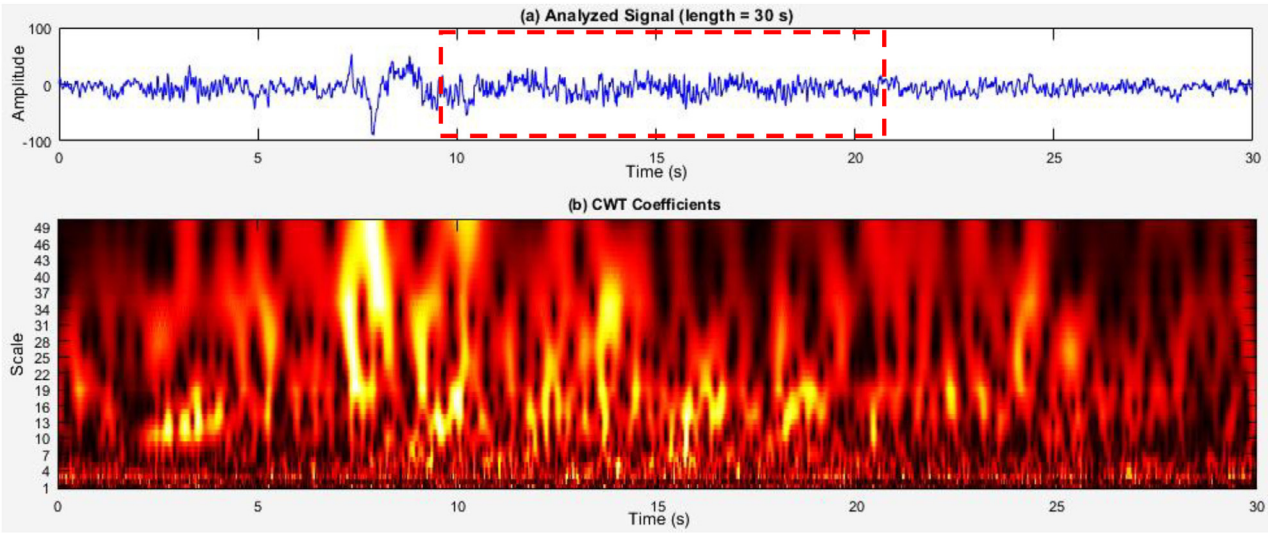

Fig. 3. (a) Duration of 30 s, 3750 samples, of EEG. The EEG arousal is shown in signal between 9 and 21 s, (b) the CWT coefficients.

frequency estimator [25]. In this study, it has been shown that the Morlet wavelet produces the smallest estimation error. The graph of the CWT coefficients is shown in Fig. 3b, wherein a remarkable change is seen in the segment that accompanies arousal. However, using the scale axis is not very suitable for observing the frequencies of arousal at a glance. The scalogram approach can be used to solve this drawback.

2.3.2. The scalogram

A scalogram is the squared magnitude of the CWT and is defined as:

$$|W_x(b,a)|^2 = \left| |a|^{-\frac{1}{2}} \int_{-\infty}^{\infty} x(t)\psi^*\left(\frac{t-b}{a}\right)dt \right|^2 \tag{3}$$

The scalogram can also be represented as images expressed with different color map or levels of gray, such as spectrograms. The use of this approach allows the result of CWT analysis to be displayed on the frequency axis instead of the scale axis. A single scale includes a frequency band. The scalogram relates the EEG arousal to peak-scale variations. The peak scale is the scale of maximum amplitude in the scalogram. The EEG arousal attenuation can be estimated from the scalogram according to the scale shift of the raw EEG signal and can also be described indirectly by the mean of a scalogram. By reason of the energy density increases, the variance of the scalogram is also associated with EEG arousal. As a result, both the mean and variance of the scalogram varies during arousal activity [23].

In Fig. 4a and b, the signal which is given in Fig. 3a and its scalogram are shown respectively. According to AASM rules, arousal is seen to be in frequency greater than 16 Hz. Since the wavelet coefficients did not change significantly in frequency values greater than 21 Hz with respect to performed experimental studies, the scalogram frequency band was selected between 16 and 21 Hz.

2.4. Feature extraction

According to experimental results of the CWT analysis, it was determined that the mean and variance values of scalogram changed remarkably where the arousal exists. For the EEG signal given in Fig. 4a, the mean and variance of the scalogram coefficients are shown in Fig. 5. Thus, the mean and variance values of the scalogram coefficients in the range of 16–21 Hz are selected as features. The SVM was used as a classifier with these features.

2.5. Classification: support vector machine

The SVM, a statistical algorithm proposed by Vladimir Vapnik in the late 1960s, was based on the principle of finding hyperplanes that can make it possible to distinguish two classes in the feature space in the most appropriate way. With an appropriate nonlinear mapping to an adequately high dimension, data from two categories can always be separated by a hyperplane [26,27]. The corresponding hyperplane can be defined as:

$$w^T x + b = 0 \tag{4}$$

Let w and b denote the weight vector and bias in the optimal hyperplane, respectively. The desired directionally geometrical distance from the sample x_i to optimal hyperplane is:

$$r = \frac{g(x_i)}{||w||} \tag{5}$$

where $g(x_i) = w^T x_i + b$ is the discriminant function as defined by the hyperplane (Fig. 6) and also called x_i's functional margin given w and b. SVM aims to find the parameters w and b for an optimal hyperplane in order to maximize the margin of separation [28]. Thus, SVM is also called maximal margin classifier. The two situations that may be encountered in SVM are the fact that the data is in a structure that can be separated linearly or non-linearly [26,27].

SVM is an important example of kernel methods from the current areas of machine learning. It is expressed as a computer algorithm, which is recognizing repeatedly occurring patterns and enhancing performance based on past experience. It enables a linear solution of a nonlinear separation problem in the sample space using the kernel function. Therefore, it is a very successful technique in generalizing complex problems such as pattern recognition, feature extraction, regression, and classification. It is often used for the classification of EEG signals and gives much better results than conventional methods [29,30]. Moreover, the effectiveness of classification is shown by applying k-fold cross-validation to SVM.

2.5.1. K-Fold cross-validation

The features were randomly grouped into K subsamples in K-fold cross-validation algorithms. K-1 subsamples were used for

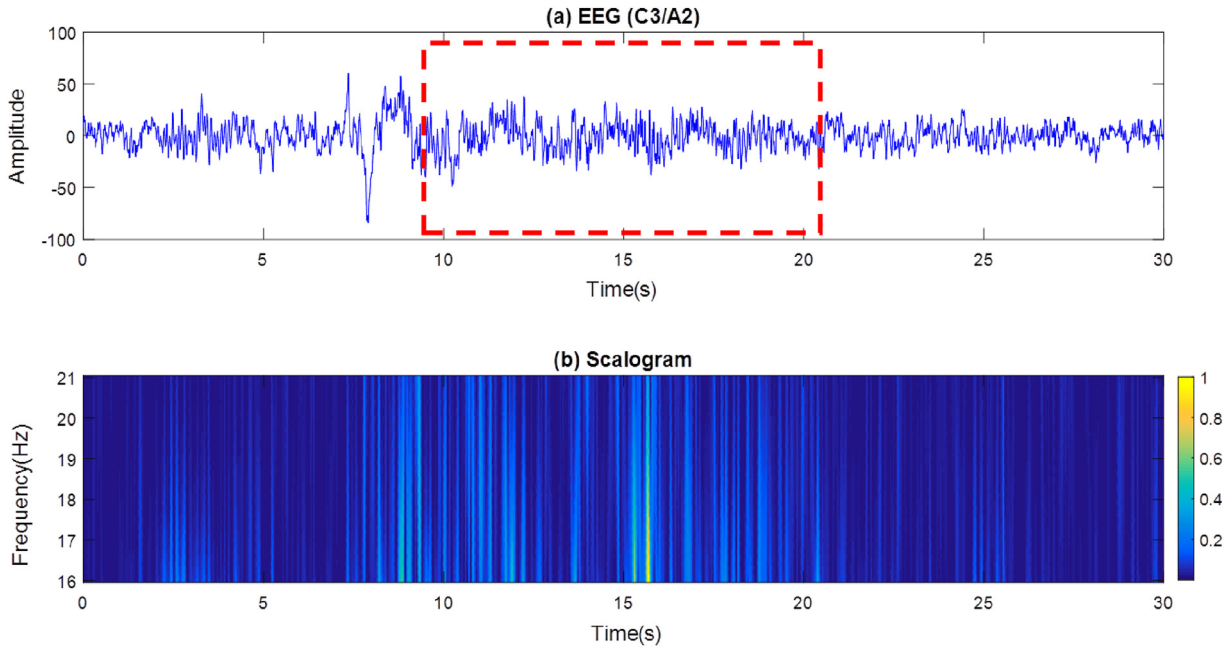

Fig. 4. (a) Duration of 30 s of EEG. The EEG arousal is shown in signal between 9 and 21 s (b) the scalogram of this epoch.

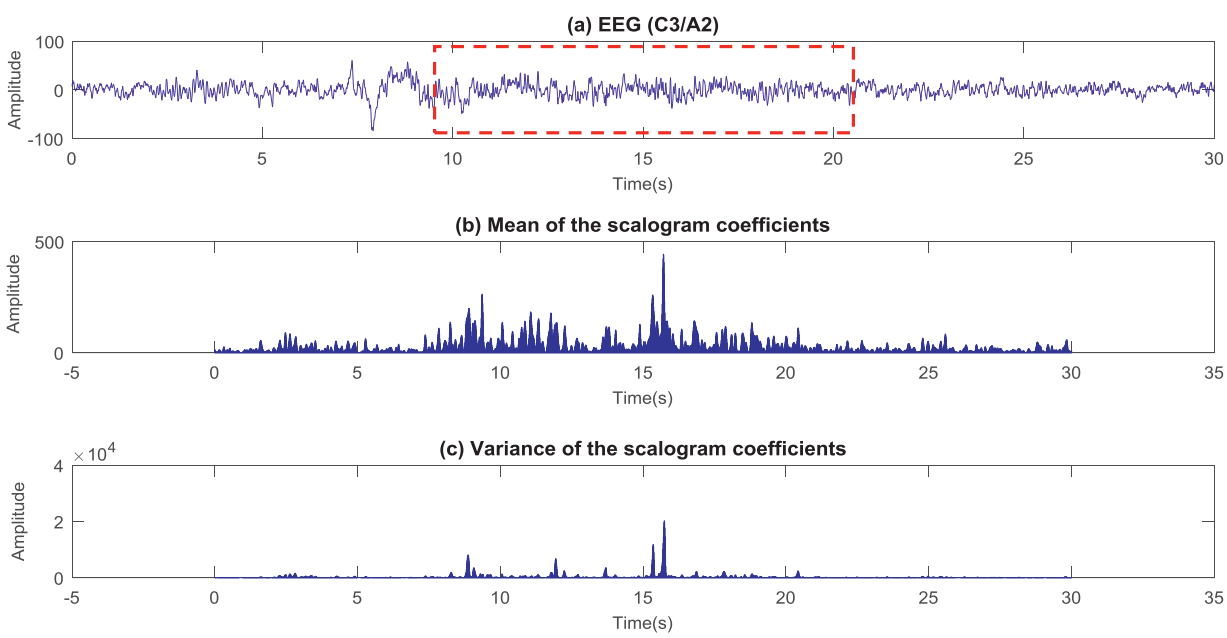

Fig. 5. (a) The EEG arousal is shown in signal between 9 and 21 s, (b) Mean of the scalogram coefficients, (c) Variance of the scalogram coefficients.

training data, and one subsample was used to test the validation data. Iterate this K times, with each of the K subsamples evaluated once. The mean of the K results was calculated from the folds to obtain a single result. All observations were therefore used for both training and validating in the K-fold cross-validation over repeated random sub-sampling [31]. In this study, the five-fold cross-validation algorithm was used.

2.6. Performance parameters calculations

All night sleep records used in the study were evaluated by physicians. The arousal scores within the scope of this evaluation were accepted as the gold standard and the success of the study was determined in this way.

The performance of the method was assessed by calculating the sensitivity, specificity, accuracy and positive predictive value (PPV). Definitions of these indices are as follows:

$$\text{Sensitivity} = \text{TP}/(\text{TP} + \text{FN}) \tag{6}$$

$$\text{Specificity} = \text{TN}/(\text{TN} + \text{FP}) \tag{7}$$

$$\text{Accuracy} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{8}$$

$$\text{Positive Predictive Value(PPV)} = \text{TP}/(\text{TP} + \text{FP}) \tag{9}$$

where true positive (TP) is the number of accurately detected actual arousal events that are scored by physician, false negative (FN) signifies the number of not found events, and false positive (FP) indicates the number of detection in case there is no arousal, and

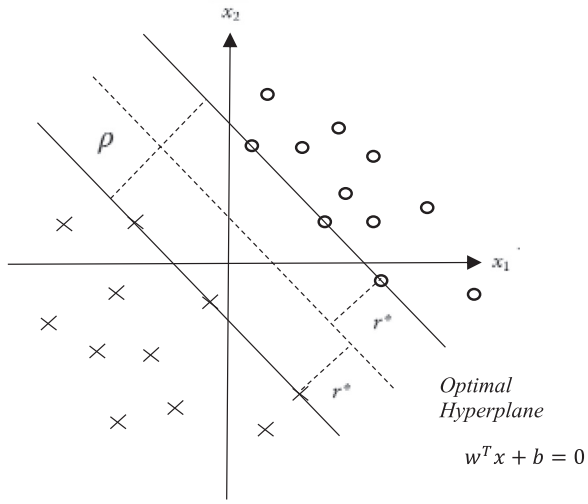

Fig. 6. Optimal hyperplane for a linearly separable case.

true negative (TN) stands for the number of correct detection while there is no arousal score [32].

3. Results

We used whole-night PSG recordings of five patients from the SHHS PSG database of NSRR. The database registration numbers of these recordings were as follows: subject1 (shhs2-203502); subject2 (shhs2-201267); subject3 (shhs2-203607); subject4 (shhs2-203634); subject5 (shhs2-203829). These records were randomly chosen out of many records in the database [17]. Since each PSG recording duration was different, 240 epochs were randomly selected from each patient record. For five subject, totally 1200 epochs were used. Fifty percent of this 1200 epoch data were used for training, the rest were used for testing. The data used in the training set were not used in the test set. In the training data set, there are arousals in 150 of the 600 epochs. In the remaining 450 epochs there is no arousal. Likewise, there are 150 arousals in the test data set and the remaining 450 epochs there is no arousal. Each record has annotations of the scored arousals (gold standard) by the sleep expert. Accordingly, we defined a binary data vector for the training and test procedure. This vector consist of arousal scores and epochs with arousal score were represented by “1”, while non-scoring epochs were represented by “0”.

In this research, various SVM architectures were trained to classify the EEG arousal. The best training result was obtained by linear SVM. In order to evaluate the performance of automatic detection, sensitivity, specificity, accuracy, and PPV were obtained. The training results are given separately in Table 1 for each patient. According to the training results, it was determined that the classifier was well trained. After that, the trained classifier was tested with unused data in the training set. Obtained test results are shown separately in Table 2 for each patient.

Table 1
Training results of SVM for five subjects.

Subject no.	Sensitivity	Specificity	Accuracy	PPV
Sub.1	76.67%	100%	94.2%	100%
Sub.2	100%	97.77%	98.3%	93.75%
Sub.3	90%	100%	97.5%	100%
Sub.4	80%	100%	95%	100%
Sub.5	96.66%	100%	99.2%	100%
Average	88.66%	99.55%	96.8%	98.75%

Table 2
Test results for five subjects.

Subject no.	Sensitivity	Specificity	Accuracy	PPV
Sub.1	83.33%	100%	95.8%	100%
Sub.2	93.33%	98.89%	97.5%	96.55%
Sub.3	93.33%	100%	98.3%	100%
Sub.4	93.33%	100%	98.3%	100%
Sub.5	100%	100%	100%	100%
Average	92.66%	99.78%	98%	99.31%

Table 3
Training and test results of all data set.

Subject 1–5	Sensitivity	Specificity	Accuracy	PPV
Train	93.33%	99.33%	97.8%	97.90%
Test	94.67%	99.33%	98.2%	97.93%

At this stage, we carried out a second training and test process to demonstrate the robustness of the algorithm. For this purpose, we have combined all the patient data previously used in the training and test sets. Thus, we used training and test data, each consisting of 600 epochs, such as single patient data. As in the previous procedure, the five-fold cross-validation algorithm was used for training and the training data were not used during the testing. Obtained results are given in Table 3. As a result, the overall sensitivity, specificity, accuracy, and positive predictive value of the algorithm are 94.67%, 99.33%, 98.2%, and 97.93%, respectively.

4. Discussion

EEG arousal that causes a discontinuity in sleep is one of the main reasons for not having a good night’s sleep. For the diagnosis of sleep disorders, accurate and complete scoring of a PSG record is required. Simultaneous analysis of multiple signal recording is a long complex task and requires professional experience in sleep medicine. Thus, automated decision support algorithms can be a valuable approach for reducing subjectivity while increasing the accuracy of scoring.

In this paper, we have proposed a new computer-aided diagnosis method for the automatic detection of EEG arousals using approaches based on time-frequency analysis and a machine learning model. We have showed that the EEG arousal pattern can be characterized by the scalogram. The features obtained from the scalogram coefficients are completely novel for the literature. The presented method needs one channel EEG and does not require any signal conditioning or pre-processing step that can cause to loss of peaks unexpectedly. This method is robust as shown here and yields reproducible results for different recordings. Moreover, no manual involvement is required during the analysis. Some of the previous studies reported include rule-based approaches, whereas others are based on a threshold value [7,11,12]. In addition, these approaches are based on imitating human visual scoring techniques. The proposed method, based on well-defined time-frequency characterization of EEG, is different from previous studies. There are no meta-rules or threshold values. Moreover, it does not directly depend on the morphology of the EEG arousal rather than unfiltered pure EEG data.

It has shown that the overall sensitivity, specificity, accuracy, and PPV of the algorithm are 94.67%, 99.33%, 98.2%, and 97.93%, respectively (Table 3). Comparison of the results of this works with the previous studies in the literature is shown in Table 4. In fact, one-to-one comparison with each study is not feasible. Because, almost every study uses its own records, and for the validation criterion the sleep physician is considered as the gold standard. For these reasons, it is difficult to make a complete comparison. However, this study gives better performance according to the results

Table 4
Comparison of the results of this work with the previous studies in the literature.

	Year	Sensitivity	Specificity	Accuracy	PPV
De Carli et al.	1999	88.1%	–	–	–
Cho et al.	2006	75.26%	93.08%	–	–
Agarwal	2006	42.7–82.2%	56.6–80.3%	–	–
Shmiel et al.	2009	75.2%	–	–	76.5%
Sugi et al.	2009	–	–	86%	–
Álvarez-Estévez et al.	2011	86%	76%	–	–
Behera et al.	2014	93.26%	91.39%	–	–
Fernández-Varela et al.	2017	74.8%	98.8%	–	–
Fernández-Varela et al.	2017	78–81%	88–89%	–	–
This study	**2019**	**94.67%**	**99.33%**	**98.2%**	**97.93%**

reported in previous studies. To minimize this benchmark problem, we used an open-access database.

As emphasized in the paper, the visualization of arousals is a time-consuming and laborious task, which requires personal expertise. Therefore, visual scoring may lead to subjective results. These evaluations might possibly be made by different physicians who specialized in different medicine discipline or visual scoring education. In fact, the same sleep expert was found to perform different scoring on the same patient data at different times [4,33]. This situation can be explained by the reasons such as fatigue and loss of concentration during visual scoring. In addition, the expertise of the physician and the professional experience are other important factors. The above-mentioned reasons are the biggest challenges in the development of computer vision-learning based automatic scoring algorithms. Therefore, inconsistency or bias in reference data, caused by physicians, result in subjective learning and biased outputs. So, the performance of automated algorithms is adversely affected and the relative success values are obtained.

5. Conclusion

In this study, a decision support algorithm, which ensures automatic detection of arousal in sleep EEG, has been developed. The proposed algorithm works with high accuracy, reproducibility and gives objective results without case-specific sensitivity. However, there are some limitations to this and similar studies. There is no standard to compare the achievements of previous studies. Because, each research group used its own data and validation criteria. In addition, the number of databases with open or limited access to PSG data that is created in conjunction with clinical sleep medicine trials is insufficient. A major deficiency in this regard is the lack of patient records with reliable scoring. We believe that the success of future works can be enhanced by the establishment of open access databases, which include independent and objective assessments that have high reliability.

Conflict of interest

None of the authors has any conflict of interest.

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